

ALPENA HARBOR, MI, USA

WMO: 997359

Lat: 45.0630N

Lon: 83.4286W

Elev: 179

StdP: 99.19

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.8	(c) -16.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.3	(k) -2.1	(l) 9.8	(m) -1.6	(n) 2.7	(o) 320	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.0	(c) 30.0	(d) N/A	(e) 27.9	(f) N/A	(g) 26.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.8	(b) 8.4	(c) 7.5	(e) -22.0	(f) 32.6	(g) 3.9	(h) 1.7	(i) -24.8	(j) 33.8	(k) -27.1	(l) 34.8	(m) -29.3	(n) 35.8	(o) -32.2	(p) 37.0		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-6.4	-6.1	-1.4	5.2	11.6	17.2	20.6	20.0	15.5	9.4	2.9	-1.6		
	DBStd	10.42	5.62	5.42	5.44	3.68	4.11	4.10	3.37	3.20	3.78	4.37	4.27	4.60		
	HDD10.0	2137	509	450	355	151	28	1	0	0	2	64	216	361		
	HDD18.3	4237	768	683	613	394	213	69	15	17	101	281	464	619		
	CDD10.0	1154	0	0	1	7	77	216	330	310	168	44	2	0		
	CDD18.3	214	0	0	0	0	4	34	87	69	17	3	0	0		
	CDH23.3	1133	0	0	0	1	42	187	482	339	54	28	0	0		
	CDH26.7	294	0	0	0	0	8	59	113	91	12	11	0	0		
(14) Wind	WSAvg	3.4	3.7	3.5	3.5	4.0	3.6	3.0	2.9	3.1	3.0	3.5	3.6	3.6		
(15) Precipitation	PrecAvg	723	42	33	52	57	74	72	73	82	77	57	56	46		
	PrecMax	895	105	99	187	105	211	213	182	150	181	166	189	113		
	PrecMin	513	4	3	9	30	25	5	6	23	7	16	16	10		
	PrecStd	105	24	21	32	19	38	42	40	36	41	32	30	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	9.7	14.0	20.7	27.7	31.6	32.8	32.5	27.7	27.0	15.0	10.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	4.7	4.8	10.9	15.5	23.9	28.2	29.8	29.2	24.6	20.5	12.0	7.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	2.7	2.5	8.3	12.7	20.6	25.3	28.0	26.8	22.6	17.3	10.2	5.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	1.3	1.1	5.6	10.5	18.0	23.0	26.2	24.9	21.1	15.5	8.7	4.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.6	7.6	7.9	7.1	8.1	7.6	8.0	8.0	7.5	6.9	5.4	5.0		
	5% DB	MCDBR	6.7	7.9	10.3	10.8	12.6	10.2	10.0	10.3	9.3	9.6	7.6	5.9		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.285	0.298	0.308	0.352	0.373	0.405	0.403	0.392	0.370	0.341	0.310	0.282		
	taud		2.438	2.371	2.449	2.373	2.389	2.323	2.348	2.376	2.430	2.489	2.522	2.493		
	Ebn at Noon		836	888	930	909	896	864	861	858	849	830	803	807		
	Edh at Noon		71	91	97	114	116	125	120	113	99	81	66	61		
(44) All-Sky Solar Radiation	RadAvg		1.25	2.17	3.68	4.65	5.52	6.03	6.10	5.17	4.04	2.42	1.34	0.92		
	RadStd		0.11	0.24	0.35	0.42	0.50	0.44	0.23	0.24	0.37	0.26	0.18	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page